



Plated wire

Round, square and flat plated wire

Luvata manufactures round, square and flat copper and copper alloy based wire, with a variety of plating options including tin and nickel as single layer or multi-layer.

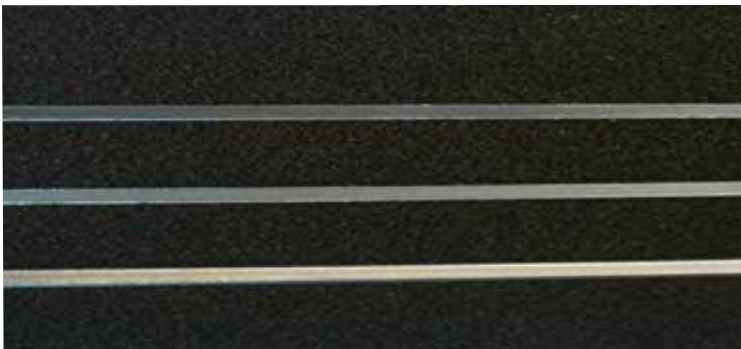
Industry applications:

- Electronics and electrical connectors
- Automotive connectors and terminals, including data connectivity systems and power distribution boxes
- E-mobility including EV production, charging infrastructure and energy storage

Advantages of Accurate Wire's electro-degreasing process

Dirt, debris and heavily oxidized areas can interrupt or hamper plating adhesion, leading to isolated defects that are detrimental in end-use applications. Other plating companies typically offer the industry standard single electro-degreasing process and often extend the use of electro-degreasing chemistries to reduce costs.

At Accurate Wire, we utilize an alkaline bipolar degreasing system with two separate bath sections dedicated to electrically activated degreasing and surface oxide removal. The system also utilizes both air wipes and a 5-step rinse between baths. The bath includes an acid activation with rapid agitation prior to each plating bath to ensure uniform surface preparation that will result in consistent plating thickness and uniform surface quality.



About Luvata

Luvata is a world leader in metal solutions manufacturing and related engineering services to industries such as renewable energy, automotive, healthcare, and power generation and distribution.

The company's continued success is attributed to its longevity, technological excellence and strategy of building partnerships beyond metals. Employing approximately 1,400 staff in 6 countries, Luvata works in partnership with customers such as ABB, CERN, Siemens and Toyota. Luvata is a group company of Mitsubishi Materials Corporation.

Base alloys

Alloy	CDA
High performance copper alloys	C151, C19210, C194, C197
Copper	C101, C102, C110
Brass	C210, C220, C230, C260
Phosphorous bronzes	C510, C511, C519, C521



Wire shapes and specification

Shape	Specifications
Round wire	0.010" to 0.100" (0.25 mm - 2.5 mm)
Square wire	0.010" to 0.0600" (0.254 mm - 1.52 mm)
Flat rectangular wire	thickness from 0.001" - 0.125" (0.025 mm - 3.2 mm) width 0.005" - 0.500" (0.125 mm - 13 mm)



Plating specifications

Material	Thickness (µm)
Nickel (Ni)	0 – 5.0 µm (0 – 200 µin)
Copper (Cu)	0 – 3.8 µm (0 – 150 µin)
Tin (Sn)	0 – 10.0 µm (0 – 400 µin)

Plating options

Single layer	nickel, copper or tin
Multi-layer	Ni-Sn, Cu-Sn, Ni-Cu-Sn (other layer configurations available upon request)



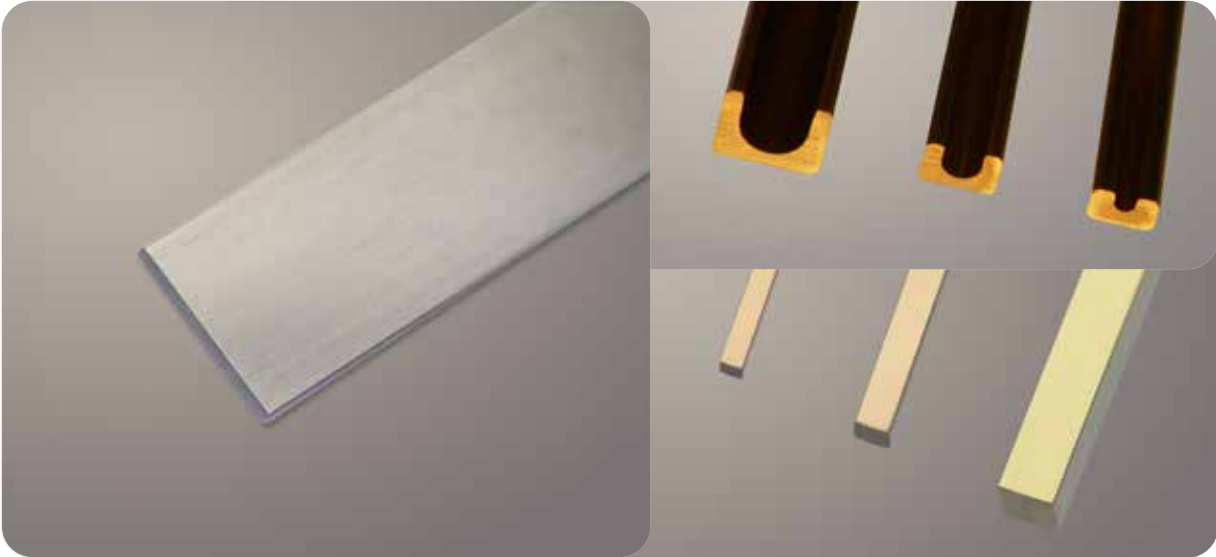
Packaging

Net spool weights range from 5 lb. to 1000 lb. (2.25 kg to 450 kg) with precision layer winding to ensure trouble-free payoff. Other packaging options available upon request.

Accurate Wire
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Branford, CT 06405 USA
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luvata.com





Accurate Wire

Square, flat and custom shaped wire

Accurate Wire was founded in 1988 and acquired by Luvata in 2016. Located in Branford, Connecticut, we specialize in manufacturing square, flat and custom shaped wire available in a wide variety of alloys. Our wire is commonly found in the aerospace, automotive, consumer products, electronics, medical and springs industries.

As a world class manufacturer of precision wire, Accurate Wire utilizes in-line laser measurement and eddy current inspection technologies to ensure defect-free wire. Our Quality Management System is certified to the ISO 9001:2015 Quality Standard.

With annealing, rolling, wire drawing and turkshead know-how, as well as expertise in designing and building our own custom equipment, Accurate Wire continues to be an industry leader in delivering high-quality specialty wires.

Accurate Wire

precision
wire shapes



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Overview

We offer precision layer winding for square and flat wire products to ensure trouble free payoff. Net spool weights range from 5 lb. to 1000 lb. (2.25 kg to 450 kg), depending on customer preference, with other spool dimensions available upon request.

Standard wire alloys and size ranges

Wire	Standard alloys	Size ranges
Square wire	brasses, phosphor-bronzes, copper and high performance copper alloys	0.0100" - 0.0600" (0.254 mm - 1.52 mm)
Flat wire (with natural round, full round or square edge)	brasses, bronzes, copper, stainless steels, carbon steels and aluminum	thickness: 0.001" - 0.125" (0.025 mm - 3.2 mm) width: 0.005" - 0.500" (0.125 mm - 13 mm)
Custom shaped wire	brasses, bronzes, copper, stainless steels, carbon steels and aluminum	contact Accurate Wire

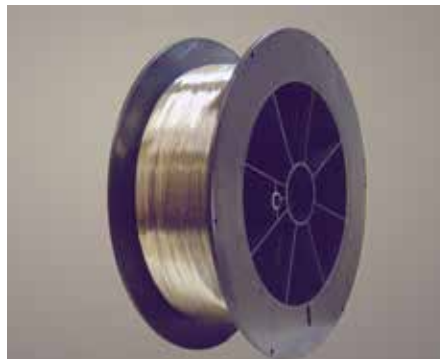
Plating specifications

Material	Thickness (µm)
Nickel (Ni)	0 – 5.0 µm (0 – 200 µin)
Copper (Cu)	0 – 3.8 µm (0 – 150 µin)
Tin (Sn)	0 – 10.0 µm (0 – 400 µin)

Plating options include single and multi-layer.

Orthodontic wire

Accurate Wire also offers precision cold rolled, square-edge orthodontic wire. We stock stainless steel type 304 vacuum melted orthodontic wire in a variety of standard sizes on 20 lb. (9 kg) spools.



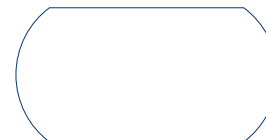
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Square wire



Flat wire



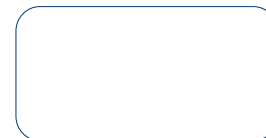
Natural Round Edge

Edge typical of rolled flat wire



Full Round Edge

Requires special tooling



Square Edge

Please specify corner radii

Flat wire edge options



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